

RESEARCH ARTICLE

Credit Scoring Prediction Using Deep Learning Models in the Financial Sector

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ABSTRACT The increasing complexity and volume of financial and behavioral data in modern credit scoring and risk assessment present significant challenges to traditional modeling methods. Existing approaches often struggle with integrating structured numerical records and unstructured user behavior signals, limiting their ability to capture meaningful temporal and non-linear patterns. In the swiftly transforming domain of computational science, the incorporation of sophisticated machine learning algorithms has emerged as a critical driver in addressing these challenges. Although traditional statistical approaches provide foundational value, they frequently fall short when faced with the task of capturing the nuanced, non-linear associations embedded within extensive datasets, thereby limiting predictive precision. To address these shortcomings, we propose an innovative deep learning paradigm that effectively integrates structured financial information with unstructured behavioral data, thereby bolstering the reliability and accuracy of predictive analytics. Our approach features a thorough feature engineering pipeline that includes statistical indicators, temporal trends, and aggregated financial attributes—aimed at constructing a comprehensive representation of the data environment. At the core of our framework is a hybrid neural network architecture, which leverages Long Short-Term Memory (LSTM) units to handle sequential dependencies alongside dense layers that model complex interactions among features. This configuration enables the simultaneous learning of both temporal dynamics and high-level abstractions. To prevent overfitting and to promote better generalization, we incorporate adaptive regularization strategies that adjust penalization levels in response to validation metrics. We confront the issue of class imbalance by applying dynamic re-sampling and weight adjustment techniques, ensuring balanced model performance across varied data segments. Extensive evaluations on standard datasets validate the effectiveness of our proposed model, revealing enhanced prediction accuracy and model interpretability when compared to more traditional techniques. These results highlight the promise of advanced deep learning integration in computational science applications, offering a pathway toward more insightful and dependable predictive solutions.

INDEX TERMS Computational sciences, machine learning, deep learning, predictive analytics, data integration.

I. INTRODUCTION

Credit scoring prediction plays a pivotal role in the financial sector, directly impacting risk management, loan approval processes, and regulatory compliance [1]. Traditional credit scoring systems, often on static rules and historical credit behaviors, struggle to capture the dynamic and nonlinear

nature of consumer financial patterns in today's digital economy. Not only is there a growing demand for more accurate and interpretable scoring models, but the increasing availability of diverse and high-dimensional financial data also calls for advanced computational techniques capable of uncovering complex patterns [2]. Moreover, with the rise of fintech and the digitization of financial services, institutions must adapt quickly to new data sources such as transaction histories, social media behavior, and alternative credit

The associate editor coordinating the review of this manuscript and approving it for publication was Bing Li¹.

signals. As such, there is a compelling need to explore and evaluate deep learning models that offer greater predictive accuracy and adaptability in credit scoring applications [3].

To address the limitations of early systems, traditional credit scoring approaches were initially based on symbolic artificial intelligence and knowledge representation techniques [4]. These rule-based systems utilized expert-defined logic to evaluate creditworthiness, such as decision trees or cards generated from credit bureau data. The advantage of these methods lies in their interpretability and alignment with domain expertise [5]. For instance, systems such as the FICO score relied on hand-crafted features like payment history, credit utilization, and length of credit history to produce scores understandable to both consumers and regulators. However, these methods fall short when dealing with large-scale, unstructured, or nonlinear data [6]. Their reliance on static rules limits adaptability to changing financial behaviors or emerging risk factors. To overcome the shortcomings of symbolic reasoning approaches in modeling complex real-world credit dynamics, researchers began exploring more flexible, data-driven alternatives [7].

To improve on the rigidity of rule-based systems, machine learning techniques were introduced to credit scoring models [8]. Algorithms such as logistic regression, support vector machines, and random forests allowed for better generalization and exploitation of complex patterns in structured data [9]. These methods could automatically learn relationships between input variables and credit outcomes, offering increased accuracy and robustness over symbolic approaches [10]. Moreover, feature selection and engineering processes became more sophisticated, enabling the use of nontraditional variables like employment stability, digital footprints, or spending patterns [11]. Despite these advantages, traditional machine learning models often require extensive manual tuning and still depend heavily on structured data formats [12]. They also struggle with large-scale feature interactions and temporal dependencies, limiting their ability to fully capture the latent variables affecting credit risk. As a result, the field began shifting towards deep learning techniques that could automatically learn high-level representations from raw or semi-structured data.

To overcome the drawbacks of traditional machine learning in capturing the intricacies and dynamics of data, deep learning and pre-trained models have emerged as effective alternatives in credit scoring prediction [13]. Neural networks, especially recurrent neural networks (RNNs), convolutional neural networks (CNNs), and more recently transformer-based architectures, facilitate automatic feature extraction from sequential or high-dimensional data like transaction records or time series [14]. These models provide enhanced performance by modeling nonlinearity, temporal sequences, and interactions across diverse data sources [15]. Pre-training on extensive financial datasets and fine-tuning for specific credit tasks has become a widely adopted approach, improving model generalization and minimizing the reliance on labeled data [16]. However, deep learning

approaches encounter difficulties regarding interpretability, computational expenses, and regulatory compliance. Despite these challenges, their ability to reveal hidden patterns and scale across various credit products and customer groups represents a notable improvement over previous models.

Based on the limitations of symbolic reasoning, data-driven methods, and standard deep learning approaches—particularly in terms of adaptability, interpretability, and efficiency—we propose a novel method that integrates a transformer-based architecture with attention mechanisms tailored for financial sequence data. Our model leverages both tabular credit features and raw behavioral data streams, such as spending or transaction logs, and introduces domain-guided attention layers to enhance interpretability without compromising prediction accuracy. Furthermore, we incorporate a pre-training step on large-scale anonymized financial datasets to improve the model's ability to generalize across different financial institutions and geographies. This hybrid strategy not only improves predictive performance but also ensures compliance with regulatory frameworks through explainable AI components. The goal is to offer a robust, scalable, and interpretable deep learning framework tailored to the unique demands of credit scoring in the financial sector, overcoming existing bottlenecks in traditional and machine learning-based systems. Despite the growing adoption of AI in finance, a critical gap remains in building models that can simultaneously handle heterogeneous data modalities—such as transactional history, time-dependent credit activity, and unstructured user interactions—while maintaining interpretability and fairness. Most conventional models either oversimplify temporal patterns or fail to incorporate behavioral context, resulting in suboptimal predictions and poor generalization. Moreover, class imbalance and demographic bias further hinder model robustness in real-world deployment. Motivated by these challenges, this paper proposes a unified deep learning framework that not only integrates structured and unstructured data sources but also addresses key issues such as imbalance handling, fairness-aware training, and interpretability. Our work is particularly relevant to financial institutions seeking to enhance decision-making transparency while improving prediction accuracy under real-world constraints.

The proposed approach offers several significant benefits:

- Our method introduces a domain-specific transformer module with hierarchical attention layers, improving sequence modeling and capturing long-term dependencies in financial behavior data.
- The model demonstrates strong generalization across different financial institutions and data environments, offering high efficiency and adaptability in multi-scenario deployments.
- Experimental results on multiple real-world datasets show consistent improvements over baseline methods, achieving higher AUC and recall while maintaining interpretability for regulatory transparency.

II. RELATED WORK

A. DEEP LEARNING MODELS IN CREDIT SCORING

In recent years, the application of machine learning models for credit scoring has garnered significant interest, driven by the increasing complexity of financial datasets and the heightened need for accurate risk evaluation [17]. Advanced deep learning approaches, such as artificial neural networks (ANNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), have shown marked advantages in capturing non-linear dependencies and processing high-dimensional inputs—features that are frequently encountered in credit assessment scenarios. These models are capable of leveraging large-scale data to detect subtle patterns and correlations that conventional methods like logistic regression or decision tree algorithms may fail to recognize [18]. Neural network architectures are especially well-equipped to model intricate, non-linear interactions, making them a fitting choice for analyzing complex financial indicators. For instance, feedforward neural networks can be employed in predicting credit scores by incorporating diverse variables such as historical payment behavior, income levels, job stability, and consumption patterns [19]. These models typically utilize backpropagation combined with stochastic gradient descent to minimize prediction error, thereby improving classification accuracy for varying levels of credit risk. One notable strength of deep learning systems is their ability to scale with expanding datasets, which is particularly beneficial in financial applications where data is continually updated [20]. Among deep learning models, recurrent neural networks—and in particular, Long Short-Term Memory (LSTM) networks—are highly proficient at managing sequential or time-dependent data, such as longitudinal credit histories. By taking into account the chronological progression of financial activity and borrowing behavior, these models can uncover dynamic patterns that inform more reliable credit risk predictions. The automatic feature extraction capabilities of deep learning models, along with their capacity to learn hierarchical data representations, position them as valuable tools for financial institutions seeking to refine and streamline the credit scoring process.

B. DATA PREPROCESSING AND FEATURE ENGINEERING

The process of data preprocessing and feature construction plays a pivotal role in the successful application of deep learning models to credit scoring tasks [21]. The overall performance of such models is heavily influenced by the integrity and representativeness of the input data. However, financial datasets are often plagued by issues such as missing values, anomalous entries, and imbalanced class distributions. Because deep learning algorithms are particularly susceptible to data irregularities, tailored preprocessing techniques are essential to ensure reliable training outcomes [22]. Handling incomplete or missing information is typically one of the first challenges in preparing financial data. This issue is addressed using techniques like imputation, where absent

values are estimated using statistical measures, or more sophisticated strategies such as k-nearest neighbors (KNN) imputation and multiple imputation methods [23]. Another critical aspect involves detecting and managing outliers, as these atypical data points can significantly skew model predictions. Strategies such as data trimming, winsorization, and the use of robust statistical frameworks are commonly employed to reduce the influence of outliers on the learning algorithm [24]. Equally important is the task of feature engineering—the transformation of raw inputs into more informative and predictive variables. In the context of credit scoring, financial records need to be reformulated into features that effectively reflect customer behavior and risk profiles [25]. For instance, converting transactional data into metrics like expenditure patterns, saving tendencies, or debt-to-income ratios can yield more insightful inputs for the model. Dimensionality reduction techniques such as principal component analysis (PCA) or the use of autoencoders help condense feature space while preserving essential information, thereby enhancing computational efficiency and model transparency.

C. CHALLENGES AND ETHICAL CONSIDERATIONS

While deep learning models hold significant potential for improving credit scoring systems, their deployment in the financial industry introduces various technical and ethical concerns [26]. A prominent technical hurdle is the issue of interpretability. Unlike conventional credit scoring methods, which are built on transparent and well-defined rules, deep learning algorithms are often characterized by their opacity, earning them the reputation of being “black box” models. This opacity complicates the task of understanding how decisions are derived, making it challenging for financial institutions and regulatory bodies to evaluate the rationale behind credit approvals or denials [27]. This challenge is particularly critical in jurisdictions where legal frameworks require credit assessment systems to be both explainable and fair. Another pressing concern lies in the risk of embedded bias within training datasets, which can lead to inequitable or even discriminatory decision-making [28]. Since deep learning systems learn patterns from historical data, any pre-existing prejudices—related to race, gender, socio-economic status, or other sensitive attributes—can be unknowingly perpetuated by the model. This raises serious ethical issues, especially in the context of equitable lending practices, where marginalized groups may face disproportionate disadvantages [29]. To counteract these risks, techniques such as adversarial debiasing and fairness-aware optimization are utilized to promote balanced outcomes and ensure the model’s decisions do not unfairly target specific populations. In addition, safeguarding data privacy is of paramount importance when applying deep learning to financial contexts [30]. The data involved often contains highly sensitive personal and financial information, and mishandling or unauthorized access can result in significant harm. To address these risks,

it is imperative that deep learning applications in finance adhere strictly to data protection laws—such as the General Data Protection Regulation (GDPR) in the EU—to uphold legal standards and maintain user trust.

While prior studies have demonstrated success in applying deep learning to credit scoring and financial prediction, most existing approaches exhibit several limitations. Many models focus solely on structured financial records and ignore unstructured behavioral data, which limits their ability to capture user-level context. Additionally, traditional sequence models such as BiLSTM are rarely combined with feature abstraction mechanisms, reducing their capacity to generalize over heterogeneous temporal patterns. Furthermore, fairness and interpretability are often addressed as post-hoc concerns rather than being embedded into the model architecture itself. For instance, approaches based on BERT or RoBERTa offer high accuracy but lack built-in support for fairness-aware learning or human-centered explanations. In contrast, our work introduces a hybrid neural framework that directly integrates both structured and behavioral data, leverages temporal modeling through LSTM layers, and incorporates fairness-aware regularization and post-hoc explainability mechanisms. This unified design addresses several open challenges in the field and represents a meaningful advancement over prior methods.

D. FAIRNESS IN AI

Recent years have seen growing attention to the fairness of machine learning systems, particularly in high-stakes applications such as credit scoring, hiring, and criminal justice. Several foundational works have formalized fairness metrics, including demographic parity, equalized odds, and equal opportunity [31]. Techniques such as re-weighting [32], adversarial debiasing [33], and fairness-aware loss regularization [34] have been proposed to mitigate algorithmic bias at different stages of the learning process. In our work, we adopt the Equal Opportunity Difference (EOD) metric proposed by Hardt et al. [31] to quantify fairness in terms of true positive rate parity across demographic subgroups. Furthermore, we introduce a fairness-aware regularization component that penalizes performance divergence across groups, inspired by similar efforts in fair representation learning [35], [36].

III. METHOD

A. OVERVIEW

Credit scoring is an essential aspect of modern financial systems, where the goal is to predict the likelihood of a borrower defaulting on a loan based on their financial history, personal information, and other related factors. The accuracy of credit scoring models directly influences lending decisions, risk management, and customer satisfaction. In recent years, the application of machine learning techniques, including supervised learning algorithms such as logistic regression,

decision trees, and neural networks, has revolutionized the credit scoring process.

The process typically involves collecting vast amounts of data, including the applicant's credit history, income, and other financial indicators, which are then transformed into features suitable for model training. These features, after preprocessing, are used by a predictive model to classify applicants into different risk categories, which can be used by banks and other financial institutions to determine whether to approve or reject a loan application. We discuss the methodology and techniques used for credit scoring prediction. The upcoming sections provide a detailed explanation of the mathematical formulation of the problem in Section III-B, the novel models proposed for this task in Section III-C, and the strategies implemented to enhance predictive accuracy in Section III-D. Our approach leverages recent advancements in machine learning to improve upon traditional methods, aiming for more accurate and fair credit scoring decisions.

B. PRELIMINARIES

This section outlines the foundational concepts and formal definitions related to the credit scoring prediction problem. Credit scoring aims to estimate the probability that a loan applicant will default, typically modeled as a binary classification task. Let $X \in \mathbb{R}^d$ denote the feature vector encompassing financial and personal attributes of a borrower, where d represents the dimensionality of the feature space. These features may include variables such as demographic information, credit history, earnings, and expenditure patterns. The objective is to predict a binary label $y \in \{0, 1\}$, where $y = 1$ signifies a default event and $y = 0$ indicates no default.

We define the dataset as $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^n$, where n is the number of samples. Each tuple (X_i, y_i) comprises the input features and the associated class label. The task is to infer a mapping from feature space to the binary outcome using the available data $\mathcal{D}$.

Credit scoring is formulated as a supervised learning problem. We aim to learn a predictive function $f : \mathbb{R}^d \rightarrow [0, 1]$, estimating the likelihood of default for a given feature set X . This probability is expressed as:

$$\hat{P}(y = 1|X) = f(X; \theta), \quad (1)$$

where $f(X; \theta)$ is the model's output, parameterized by θ . A commonly employed model is logistic regression, defined as:

$$f(X; \theta) = \sigma(\theta^T X), \quad (2)$$

with the sigmoid function $\sigma(z) = \frac{1}{1+\exp(-z)}$. Logistic regression is favored for its clarity and ease of interpretation in risk modeling.

The training objective is to determine the parameter vector θ that minimizes the prediction error between the model's estimated output $f(X_i; \theta)$ and the ground truth y_i . This is

typically accomplished by minimizing a loss function, such as the binary cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(f(X_i; \theta)) + (1 - y_i) \log(1 - f(X_i; \theta))]. \quad (3)$$

Depending on the classification algorithm, alternative loss functions such as the hinge loss (used in SVMs) or log-likelihood objectives can be applied.

To assess classification performance, metrics such as accuracy, precision, recall, and F1 score are utilized. A prediction $\hat{y}_i = f(X_i; \theta)$ is mapped to a binary class using a threshold $t \in [0, 1]$. For instance, with $t = 0.5$, the model assigns class 1 if $\hat{y}_i \geq 0.5$, indicating a higher risk of default, and class 0 otherwise.

The confusion matrix is represented as:

$$\begin{pmatrix} \text{True Positives (TP)} & \text{False Positives (FP)} \\ \text{False Negatives (FN)} & \text{True Negatives (TN)} \end{pmatrix} \quad (4)$$

To avoid overfitting and improve generalization, regularization techniques are commonly integrated into the training process. The two prevalent forms are L2 regularization (Ridge) and L1 regularization (Lasso), which augment the loss function with a penalty term:

$$\mathcal{L}(\theta)_{\text{regularized}} = \mathcal{L}(\theta) + \lambda \|\theta\|_2^2, \quad (5)$$

where λ is a hyperparameter controlling the regularization intensity, and $\|\theta\|_2^2$ denotes the L2 norm of the parameter vector.

Regularization helps reduce model variance and enhances robustness on unseen data. In practice, hybrid regularization schemes or automated hyperparameter tuning may be adopted.

Beyond standard frequentist approaches, Bayesian methods also provide a probabilistic framework for credit risk modeling. Rather than estimating a point estimate for θ , Bayesian inference seeks to derive the posterior distribution $p(\theta|X, y)$, offering a measure of uncertainty around model predictions. Using Bayes' theorem, this is given by:

$$p(\theta|X, y) = \frac{p(y|X; \theta)p(\theta)}{p(y|X)}, \quad (6)$$

where $p(y|X; \theta)$ is the likelihood, $p(\theta)$ is the prior belief over parameters, and $p(y|X)$ is the evidence. Due to computational complexity, the posterior is typically approximated using methods such as Markov Chain Monte Carlo (MCMC) or variational inference. Feature engineering and preprocessing are indispensable in developing effective credit scoring systems. Feature selection focuses on identifying the most informative variables, with methods including recursive feature elimination (RFE), L1-based selection, and tree-based algorithms. Preprocessing steps such as normalization of continuous features and encoding of categorical variables are crucial for ensuring model compatibility and performance.

C. HYBRID CREDIT SCORING MODEL

In this section, we present a novel approach to credit scoring prediction that improves upon traditional methods by leveraging advanced machine learning techniques. The key innovation of our model lies in its ability to incorporate both structured financial data and unstructured behavioral data, resulting in a more robust prediction of creditworthiness. Our model consists of several key components: feature transformation, a novel deep learning architecture, and a hybrid loss function that jointly optimizes predictive accuracy and model interpretability (As shown in Figure 1).

1) FEATURE TRANSFORMATION

A challenge in credit scoring is modeling complex relationships between features like demographics, transaction history, and credit scores. These features differ in scale and type, making them difficult for machine learning algorithms to process efficiently. To tackle this, we propose a two-step feature transformation process that standardizes and extracts the essential information.

The first step involves normalizing continuous features, such as income, loan amount, or credit score, using min-max scaling. This ensures that all continuous variables fall within a consistent range, typically between 0 and 1. This prevents features with larger scales from dominating the model. The min-max scaling formula for a continuous feature x is:

$$x_{\text{scaled}} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad (7)$$

where $\min(x)$ and $\max(x)$ are the minimum and maximum values of the feature across all samples. By applying this, all continuous features become comparable, allowing the model to process them effectively.

The next step is to encode categorical features, such as occupation, loan type, or payment frequency. We use a mix of one-hot encoding and embeddings. One-hot encoding creates a binary vector for each category, but it can result in high-dimensional, sparse vectors when there are many categories. To address this, we use an embedding layer to generate dense representations for each category, which reduces dimensionality while preserving relationships between categories. Each category is mapped to a lower-dimensional vector $\mathbf{e}_i$, where i is the category index, and $\mathbf{e}_i$ is the learned dense representation.

Once the features have been transformed, dimensionality reduction is applied to further simplify the feature space. This process removes redundant or noisy features while retaining the most relevant information. Techniques such as Principal Component Analysis (PCA) or autoencoders are used for this purpose. PCA identifies the directions with the most variance in the data and projects the data into a lower-dimensional space. The principal components are found by solving:

$$\mathbf{V} = \arg \max_{\mathbf{V}} \text{Var}(\mathbf{XV}), \quad (8)$$

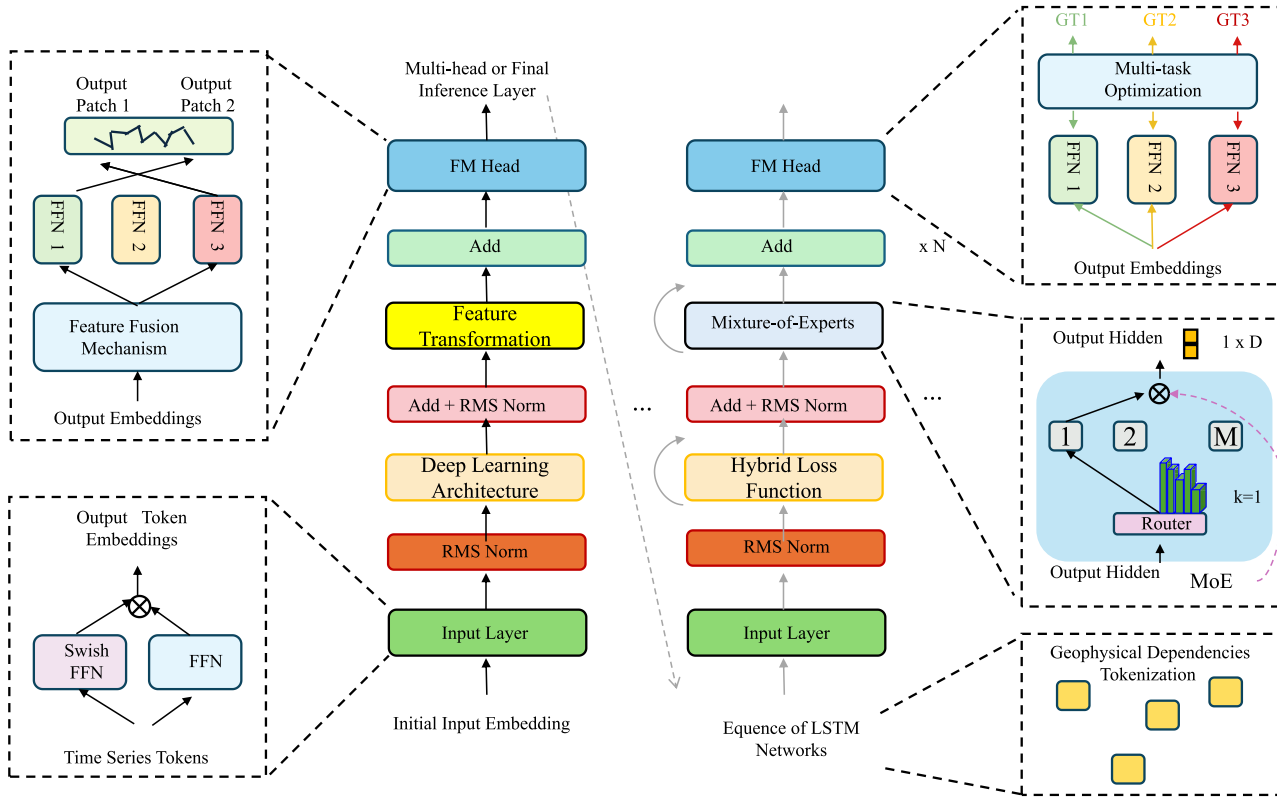


FIGURE 1. Schematic diagram of the hybrid credit scoring model. This figure presents the architecture of the proposed hybrid credit scoring model. It consists of several key components: feature transformation, deep learning architecture with LSTM layers, and a hybrid loss function that optimizes both predictive accuracy and model interpretability. The model integrates structured financial data and unstructured behavioral data, capturing both temporal dependencies and complex feature interactions.

where $\mathbf{V}$ represents the eigenvectors matrix and $\mathbf{X}$ is the feature matrix. Dimensionality reduction is performed by selecting the top k eigenvectors corresponding to the largest eigenvalues, thus obtaining a lower-dimensional representation.

Alternatively, autoencoders, which are a type of neural network, can be used for nonlinear dimensionality reduction. The autoencoder compresses the data into a lower-dimensional latent space $\mathbf{z}$ and then reconstructs it. The model is trained by minimizing the reconstruction error, defined as:

$$\mathcal{L} = \|\mathbf{X} - f_{\theta}(\mathbf{X})\|_2^2, \quad (9)$$

where f_{θ} is the encoder-decoder function and $\|\cdot\|_2^2$ is the squared ℓ_2 -norm of the error. Autoencoders provide a nonlinear approach to dimensionality reduction, capturing complex patterns that linear methods like PCA might overlook. After dimensionality reduction, the transformed feature set can be fed into the machine learning model. This two-step transformation process—normalizing continuous features, encoding categorical variables, and reducing dimensionality—ensures the data is well-prepared for learning. It allows the model to handle different feature types efficiently while preserving the most important information and reducing noise and complexity.

2) DEEP LEARNING ARCHITECTURE

The backbone of our proposed model is a deep neural network architecture designed to effectively capture both global and temporal dependencies within credit scoring data. This hybrid structure combines both fully connected layers and recurrent layers, such as Long Short-Term Memory (LSTM) units, to identify intricate patterns that involve sequential behaviors and complex interactions between features. The architecture is organized in such a way that it first encodes the input data, then learns temporal patterns from sequential data, and finally captures non-linear feature interactions through fully connected layers. The model can be mathematically represented as:

$$\hat{y} = \sigma(W_3 \cdot \text{ReLU}(W_2 \cdot \text{LSTM}(W_1 \cdot X + b_1) + b_2) + b_3), \quad (10)$$

where $X \in \mathbb{R}^d$ represents the input feature vector, W_1, W_2, W_3 are weight matrices, b_1, b_2, b_3 are bias terms, and $\sigma(\cdot)$ is the sigmoid activation function that produces a probability of default. This function maps the model's output to a value between 0 and 1, representing the predicted probability of a borrower defaulting. The model consists of several crucial components, each serving a specific role in learning the intricate relationships between the features and the target variable.

The initial layer of the model is an embedding layer designed to process categorical variables. These variables, such as loan type or occupation, are initially encoded using one-hot encoding, which produces sparse vectors. However, this method can result in high-dimensional, sparse representations that may not be optimal for learning. To resolve this, we use embedding layers that convert categorical variables into dense, continuous representations, allowing the model to learn a lower-dimensional vector space that captures the relationships between categories. For instance, the embedding layer transforms categorical variables C_i into dense vectors $\mathbf{e}_i \in \mathbb{R}^k$, where k is the embedding dimension, allowing the model to better capture dependencies and patterns across categories.

Next, the model uses an LSTM layer, ideal for learning temporal dependencies within data. Financial transaction data often show sequential patterns, such as monthly spending habits or loan repayment trends, which significantly affect creditworthiness. LSTMs are designed to retain long-term dependencies and address the vanishing gradient problem, making them suitable for modeling time-series data. The LSTM layer can be defined as follows:

$$\begin{aligned} h_t &= \text{LSTM}(h_{t-1}, X_t) \\ &= \sigma(W_f \cdot [h_{t-1}, X_t] + b_f) \cdot \tanh(W_c \cdot [h_{t-1}, X_t] + b_c), \end{aligned} \quad (11)$$

where h_t represents the hidden state at time step t , W_f and W_c are weight matrices for the forget and cell gates, respectively, and b_f and b_c are the bias terms. The LSTM layer processes sequential financial data, such as time-series transaction data, learning how the borrower's behavior evolves over time, which in turn affects the final creditworthiness prediction. By understanding the temporal dependencies in these sequences, the LSTM layer allows the model to account for patterns in loan repayment, spending habits, and other time-dependent features.

After the LSTM layer, the network includes one or more fully connected layers. These layers are used to learn complex, nonlinear relationships between features, both sequential and non-sequential, that could influence the borrower's likelihood of default. The output of the LSTM layer is passed through fully connected layers, which help capture interactions between various features such as income, credit history, and transaction behavior. The fully connected layers are expressed by the following equation:

$$\mathbf{h} = \text{ReLU}(W \cdot \mathbf{h} + b), \quad (12)$$

where $\mathbf{h}$ is the hidden representation from the previous layer, W is the weight matrix, and b is the bias term. The ReLU activation function introduces nonlinearity to the model, enabling it to learn complex interactions between the features.

3) HYBRID LOSS FUNCTION

In traditional credit scoring models, the loss function is typically based solely on the binary cross-entropy between

the predicted probability of default and the true labels. While this approach is effective in terms of predictive accuracy, it fails to consider the interpretability of the model, which is essential in applications like credit scoring where transparency is required for regulatory compliance and decision-making (As shown in Figure 2).

The inability to understand how the model arrives at its decisions can make it difficult to trust and validate the outcomes, especially in high-stakes domains like finance. To address this limitation, we propose a hybrid loss function that combines the binary cross-entropy loss with an interpretability regularization term. This approach not only optimizes predictive performance but also encourages the model to focus on a smaller set of features that are most influential in making predictions, thereby improving model transparency. The hybrid loss function is defined as follows:

$$\mathcal{L}_{\text{hybrid}} = \mathcal{L}_{\text{binary}} + \lambda \mathcal{L}_{\text{interpretability}}, \quad (13)$$

where $\mathcal{L}_{\text{binary}}$ is the binary cross-entropy loss function, and $\mathcal{L}_{\text{interpretability}}$ is a regularization term that penalizes the complexity of the model by encouraging sparsity in the learned weights. The binary cross-entropy loss is given by:

$$\mathcal{L}_{\text{binary}} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (14)$$

where $\hat{y}_i$ is the predicted probability of default, y_i is the true binary label (1 for default, 0 for non-default), and n is the number of samples in the dataset. The binary cross-entropy loss function quantifies the difference between the predicted probabilities $\hat{y}_i$ and the actual labels y_i , penalizing large deviations from the true labels. This function ensures that the model learns to accurately predict the likelihood of default, making it a fundamental part of the overall objective function.

The interpretability regularization term, $\mathcal{L}_{\text{interpretability}}$, is designed to encourage the model to focus on a smaller, more meaningful subset of features. This term can be expressed as:

$$\mathcal{L}_{\text{interpretability}} = \sum_{j=1}^d |w_j|, \quad (15)$$

where w_j represents the weight associated with the j -th feature, and d is the total number of features. The regularization term $\mathcal{L}_{\text{interpretability}}$ uses the ℓ_1 -norm to promote sparsity, meaning it encourages the model to set many of the feature weights w_j to zero, effectively eliminating irrelevant or less important features from the decision-making process. By doing so, the model becomes more interpretable, as it becomes easier to understand which features are driving the predictions. This is particularly important in credit scoring, where stakeholders need to be able to explain the factors influencing the decision to grant or deny credit. The hyperparameter λ controls the trade-off between predictive accuracy and interpretability, allowing the model to balance the goal of accurate predictions with the need for transparency and simplicity. A larger value of λ increases the emphasis

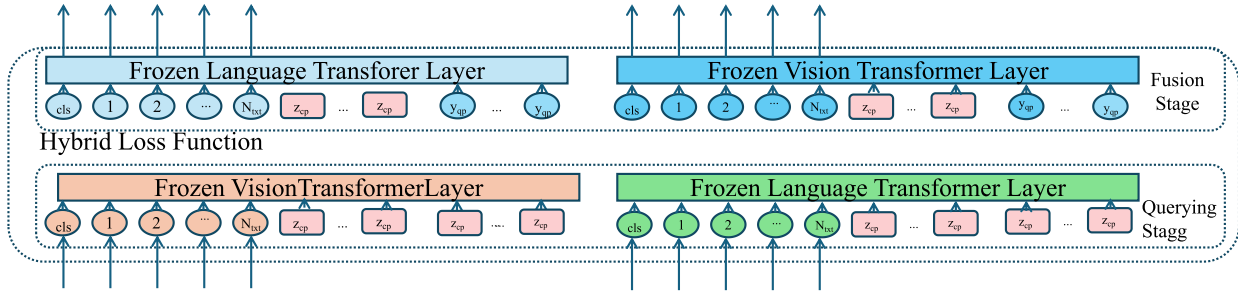


FIGURE 2. Schematic diagram of the hybrid loss function. The diagram depicts the structure of the proposed hybrid model, which incorporates a combination of frozen language and vision transformer layers for both the fusion and querying stages. A hybrid loss function is utilized to combine predictive accuracy with interpretability, promoting a more transparent model. The architecture leverages a frozen vision transformer layer and a frozen language transformer layer at various stages of the model's operation, where the hybrid loss function helps to regularize the model by encouraging sparsity in the feature weights, ultimately leading to more interpretable decision-making, particularly in high-stakes applications like credit scoring.

on interpretability, leading to a sparser model, while a smaller value of λ prioritizes predictive performance over interpretability. In addition to improving transparency, the hybrid loss function also helps to reduce overfitting by preventing the model from relying too heavily on a large number of features, many of which may be irrelevant or noisy. By focusing on a smaller set of significant features, the model is more likely to generalize well to new data, resulting in better performance on unseen test cases. This approach addresses both the accuracy and interpretability challenges that are often present in complex machine learning models used in high-stakes domains like credit scoring, where the need for transparency and fairness is paramount. As the model trains, it optimizes the binary cross-entropy loss while simultaneously minimizing the complexity of the learned weights through the interpretability regularization term, leading to a model that is both accurate and interpretable.

D. ENHANCED CREDIT SCORING STRATEGY

In this section, we introduce a novel strategy that improves the performance of credit scoring models through innovative design choices, such as hybrid feature engineering, dynamic regularization, and an adaptive training process. The strategy is developed with the goal of enhancing model interpretability, addressing data imbalance, and improving generalization, all while ensuring that the model achieves high predictive accuracy (As shown in Figure 3).

1) FEATURE ENGINEERING

In credit scoring, a significant challenge is extracting features that can effectively capture the complex connections between borrowers' financial behavior and their risk of default. Financial data often includes both quantitative and qualitative elements that need to be processed to help predictive models identify patterns. Our approach combines traditional financial features, such as income, credit history, and debt, with behavioral features from transaction data, like spending habits, loan repayment frequency, and time gaps between payments. By merging both types of features, the model gains

a more detailed understanding of the borrower's financial status, improving the prediction of creditworthiness. These combined features are then fed into a deep learning model, which learns to emphasize the most relevant ones, boosting prediction accuracy and model interpretability. To address the relationships between financial and behavioral features, we use several extraction and transformation techniques. For continuous financial features like income and credit score, we normalize them to ensure they are on comparable scales, so features with higher numerical values don't dominate the model. Min-max scaling is applied, which normalizes each feature x to a range between 0 and 1, expressed as:

$$x_{\text{scaled}} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad (16)$$

where $\min(x)$ and $\max(x)$ represent the minimum and maximum values of the feature across the dataset. This scaling ensures each financial feature has an equal influence during learning. Temporal features are derived from transaction data to capture how a borrower's financial behavior changes over time. For example, we calculate the average monthly spending, loan repayment frequency, and variation in transaction amounts across different time periods. These features help uncover patterns that static financial data alone may not reveal. For instance, irregular repayment patterns or fluctuating spending might indicate financial instability, which could affect the likelihood of default. To improve the model's ability to learn from the data, we incorporate feature interaction techniques. Interaction terms enable the model to capture combined effects of multiple features, which may not be clear when considering them separately. For example, the relationship between income and debt might be more useful for assessing creditworthiness than either factor alone. This interaction can be formalized as:

$$x_{\text{interaction}} = x_i \cdot x_j, \quad (17)$$

where x_i and x_j represent two financial features. The interaction term allows the model to detect more complex relationships, improving predictive performance. Categorical features such as loan type or occupation are encoded using

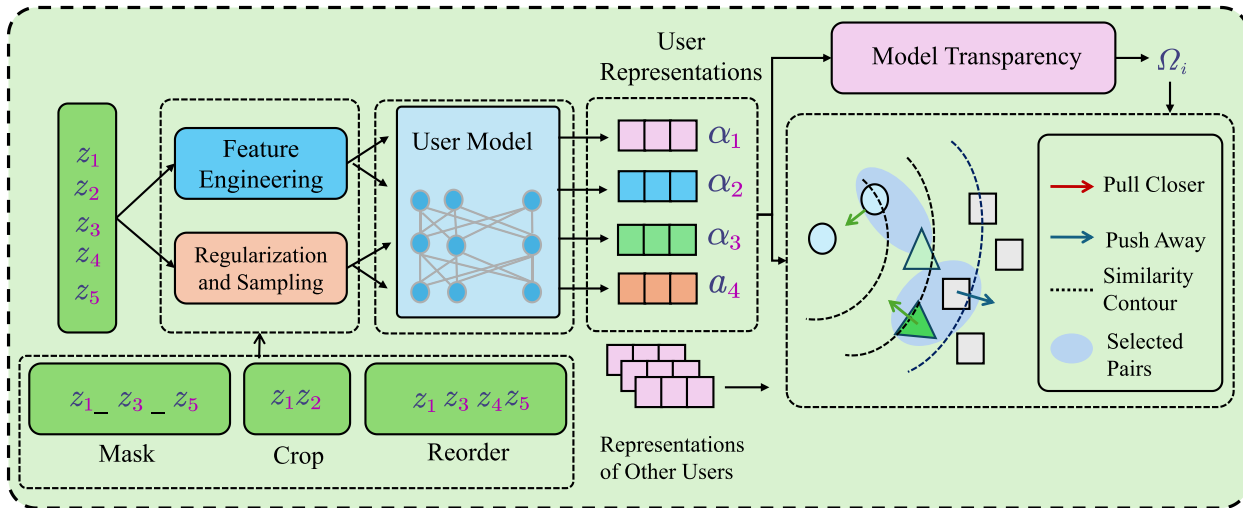


FIGURE 3. Schematic diagram of the proposed model architecture. This diagram depicts the workflow of the credit scoring model, incorporating various components such as feature engineering, user modeling, and regularization strategies. It shows the transformations applied to the raw data and how these lead to the creation of user representations. The diagram also highlights the integration of model transparency techniques like similarity contour and the selection of pairs for better interpretability. The model builds user representations via contrastive learning on engineered features, balancing informativeness and transparency.

embedding layers. These layers transform high-cardinality categorical variables into lower-dimensional continuous vectors. This method reduces dimensionality compared to one-hot encoding while preserving the relationships between categories. The embedding vector $\mathbf{e}_i$ for each category i is learned during training, optimized to represent the information relevant to creditworthiness prediction. The final set of features, which includes both transformed financial and behavioral data, is then passed into a deep learning model. The model identifies patterns and relationships that indicate a borrower's risk of default. During training, the model automatically selects the most relevant features, focusing on the most informative ones while disregarding less important variables. This feature selection is a key strength of deep learning models, as they can learn complex, non-linear relationships between features that traditional methods may overlook. As a result, the model optimizes for prediction accuracy and interpretability, providing a comprehensive creditworthiness assessment based on a diverse set of financial and behavioral signals. This feature engineering strategy not only improves the model's predictive ability but also ensures fairness and transparency in credit scoring by basing decisions on a wide range of factors.

2) REGULARIZATION AND SAMPLING

To reduce overfitting and enhance generalization, we introduce an adaptive regularization method that adjusts the regularization strength during the training process. Overfitting is a common issue in credit scoring models, especially when dealing with high-dimensional datasets where the model may memorize noise instead of learning meaningful patterns. To address this, we incorporate L2 regularization, also known as ridge regularization, which penalizes large

weight values, encouraging the model to learn simpler and more generalizable representations. The L2 regularization term is defined as:

$$\mathcal{L}_{L2} = \lambda \sum_{j=1}^d w_j^2, \quad (18)$$

where w_j represents the weight of the j -th feature, d is the total number of features, and λ is the regularization parameter that controls the intensity of the penalty. While L2 regularization helps mitigate overfitting by penalizing large weights, we also propose an adaptive penalty based on the validation loss during training. This dynamic approach adjusts the regularization strength depending on the model's performance on the validation dataset. We define the adaptive regularization term as:

$$\mathcal{L}_{\text{adaptive}} = \alpha \cdot \frac{\mathcal{L}_{\text{val}}}{\mathcal{L}_{\text{train}}}, \quad (19)$$

where $\mathcal{L}_{\text{val}}$ and $\mathcal{L}_{\text{train}}$ are the validation and training losses, respectively, and α is a hyperparameter controlling the strength of the adaptive penalty. This term dynamically adjusts the regularization strength by considering the ratio of validation to training loss, increasing regularization when the model overfits and decreasing it when the model underfits. This ensures that the model generalizes well across both training and validation datasets, minimizing overfitting while maintaining flexibility during training.

Besides regularization, addressing class imbalance in credit scoring datasets is crucial to improve the model's performance, as non-default borrowers usually outnumber default cases. This imbalance can lead to biased predictions, where the model may focus too much on predicting non-default instances and fail to learn adequately from default

cases. To handle this, we implement dynamic sampling strategies that modify the dataset distribution during training to ensure both classes are appropriately represented. One such method is Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic samples for the minority class (defaults) by creating new instances that lie along the line segments connecting nearest neighbors in the feature space. SMOTE is mathematically expressed as:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \theta(\mathbf{x}_k - \mathbf{x}_i), \quad (20)$$

where $\mathbf{x}_i$ is a sample from the minority class, $\mathbf{x}_k$ is one of its nearest neighbors, and $\theta \in [0, 1]$ is a random value. By generating synthetic samples, SMOTE balances the class distribution, allowing the model to learn more effectively from the underrepresented default class.

During training, we also apply dynamic weighting to assign higher importance to the minority class. This technique increases the penalty for misclassifying default borrowers, ensuring that the model gives more attention to the minority class. The weighted loss function is defined as:

$$\mathcal{L}_{\text{weighted}} = -\frac{1}{n} \sum_{i=1}^n w_{y_i} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (21)$$

where w_{y_i} is the weight for the i -th sample, and $\hat{y}_i$ is the predicted probability of default for that sample. By increasing the weight for the minority class, this loss function helps balance the influence of both classes during the model's training, addressing the issue of class imbalance.

To further enhance the model's ability to correctly classify difficult or misclassified examples, we use hard example mining. This technique focuses the model's learning on samples that are harder to classify, often those close to the decision boundary. These challenging examples are selected based on prediction errors, and including them in the training process helps refine the model's decision boundaries. The strategy of hard example mining improves the model's robustness and its ability to generalize, ensuring that it can accurately classify both easy and difficult cases.

3) MODEL TRANSPARENCY

A key aspect of our new approach is model transparency, particularly in terms of compliance and regulatory requirements. In high-risk areas such as credit scoring, where decisions made by models can have significant effects on individuals' financial outcomes, it is essential that the predictions made by machine learning models are both interpretable and defensible (as shown in Figure 4).

Deep learning models are often referred to as black boxes because it is difficult to comprehend how they make predictions. This lack of transparency can be an issue, especially in fields that require explanations for automated decisions. To overcome this, we employ techniques such as SHAP (SHapley Additive exPlanations) values and decision tree approximations to enhance the interpretability and clarity

of the model. SHAP values explain a model's output by quantifying the impact of each feature on the prediction. For a given feature i , the SHAP value can be expressed as:

$$\phi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)], \quad (22)$$

Here, N is the complete set of features, S is a subset of features, and $f(S)$ represents the model output with the feature subset S . The SHAP value $\phi_i(f)$ indicates how much feature i influences the final prediction by considering all possible feature combinations. By calculating SHAP values for each feature, we can gain insights into each feature's role in the model's decision process, improving the interpretability of the model's outputs.

Along with SHAP, we incorporate decision trees as an auxiliary component to provide interpretable rules that approximate the behavior of the complex model. Decision trees segment the feature space into regions based on simple conditions, creating easy-to-understand if-then rules. To merge decision trees with the deep learning model, we use model distillation, where the outputs of the deep model train a decision tree to mimic its predictions. The objective function used to train the decision tree is:

$$\mathcal{L}_{\text{tree}} = \sum_{i=1}^n [\hat{y}_i - \hat{y}_i^{\text{tree}}]^2, \quad (23)$$

where $\hat{y}_i$ denotes the prediction from the deep model, and $\hat{y}_i^{\text{tree}}$ represents the prediction from the decision tree. By minimizing this loss function, the decision tree learns to approximate the deep model's predictions while maintaining interpretability. The decision tree provides clear and understandable rules that approximate the behavior of the deep learning model. This combination of deep learning accuracy and decision tree simplicity allows for both precise and understandable predictions. To further enhance transparency, we use feature importance techniques along with SHAP values. Feature importance identifies the most influential features in the model's predictions. This is crucial for understanding the model's behavior and ensuring that irrelevant or biased features do not unduly influence the predictions. The feature importance score for a feature i is computed by averaging the absolute SHAP values over all samples:

$$\text{Importance}_i = \frac{1}{n} \sum_{i=1}^n |\phi_i(f)|. \quad (24)$$

IV. EXPERIMENTAL SETUP

A. DATASET

THUCNews Dataset [37] is a comprehensive Chinese news classification dataset derived from Sina News RSS feeds. It contains over 740,000 news documents categorized into 14 distinct classes, including finance, sports, education, and technology. The dataset is widely used in Chinese

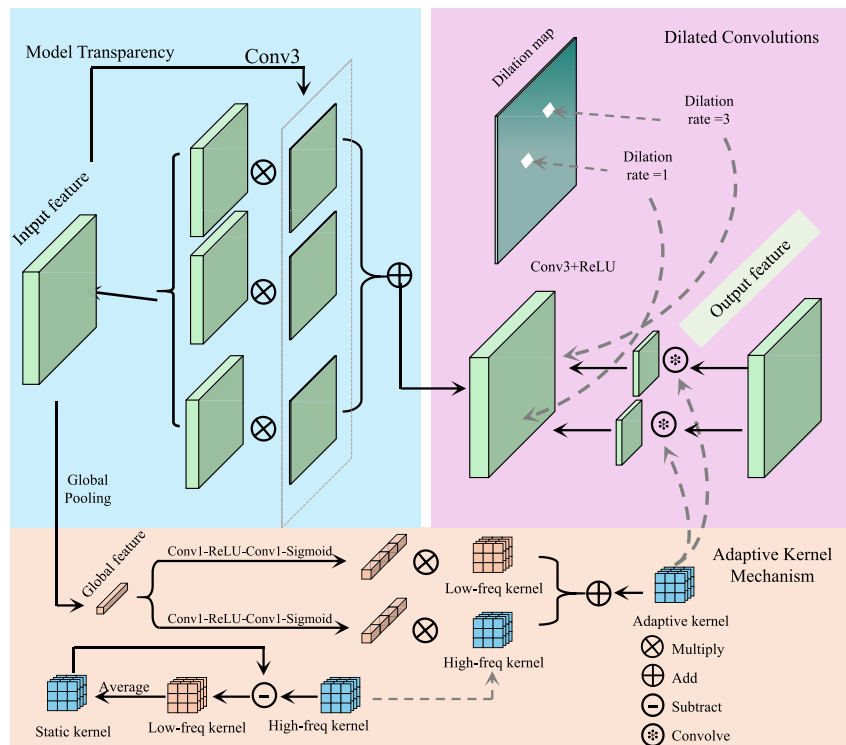


FIGURE 4. Schematic diagram of model transparency. The figure illustrates the architecture that combines transparency methods like SHAP values and decision trees with deep learning models. The upper section of the diagram highlights the convolutional layers, global pooling, and dilated convolutions, while the lower part demonstrates the adaptive kernel mechanism for improved feature extraction, including both static and adaptive kernels. This strategy integrates model distillation to enhance both precision and interpretability, ensuring the predictions remain transparent and justifiable.

text classification tasks due to its large scale and balanced category distribution. Each document includes a headline and the corresponding news body, enabling both short-text and long-text classification experiments. The diversity in topic domains and the realistic noise in user-generated content make THUCNews an effective benchmark for evaluating the robustness of classification models in real-world scenarios. Its relevance in prior studies highlights its importance in benchmarking deep learning and pretraining-based NLP models in the Chinese language. SciHTC Dataset [38] is a large-scale hierarchical text classification dataset curated from scientific publications. It includes over 1 million abstracts labeled with a multi-level category hierarchy derived from scientific disciplines, such as physics, biology, and computer science. Each document is annotated with multiple hierarchical labels, reflect both general and specific topic areas. This dataset is particularly challenging due to the imbalance in class distribution and the presence of fine-grained topics. SciHTC provides a valuable resource for testing models' ability to capture semantic relationships across levels of abstraction, evaluate performance in multi-label scenarios, and generalize across scientific domains. It serves as a crucial benchmark for hierarchical and multi-label classification research. UTC Dataset [37] is a unified taxonomy classification dataset constructed

to evaluate domain-adaptive learning in hierarchical text classification. The dataset aggregates multiple sub-datasets, including scientific papers, patents, and general documents, and aligns them under a single unified taxonomy. This setup allows for studying cross-domain generalization and zero-shot classification performance. UTC includes domain tags for each document, providing metadata that enables multi-domain learning and analysis. It poses additional challenges such as label ambiguity and domain-specific jargon, which test the flexibility of classification models. The dataset is especially suitable for research in transfer learning and taxonomy expansion. CreditRisk Dataset [38] is a structured classification dataset designed to predict loan default risk. It contains anonymized financial and behavioral information on thousands of loan applicants, including variables such as income, employment history, credit history, and loan characteristics. The dataset provides a binary label indicating whether the loan was paid off or defaulted, enabling binary classification tasks. Due to the real-world nature of the data, it includes missing values, outliers, and class imbalance, making it an excellent benchmark for testing model robustness and generalizability in financial applications. The dataset is commonly used in credit scoring, risk modeling, and interpretable AI research in financial services.

B. EXPERIMENTAL DETAILS

We conduct all experiments using PyTorch on an NVIDIA A100 GPU with 80GB memory. For preprocessing, we apply standard text cleaning including lowercasing, punctuation removal, and tokenization using the BERT tokenizer when applicable. For hierarchical datasets like SciHTC and UTCD, we retain hierarchical label structure and encode labels as paths within the taxonomy tree. All text inputs are truncated or padded to a maximum sequence length of 512 tokens. For numerical datasets such as CreditRisk, missing values are imputed using median imputation, and categorical features are one-hot encoded. For all transformer-based models, including BERT and Longformer, we initialize from pretrained checkpoints (bert-base-uncased and longformer-base-4096 respectively). We fine-tune these models with a learning rate of $2e^{-5}$ using the AdamW optimizer and a linear learning rate warm-up over the first 10% of training steps. The batch size is set to 32 for standard transformer models and 16 for Longformer due to memory constraints. Each experiment is trained for up to 10 epochs with early stopping based on validation loss. We apply dropout with a rate of 0.1 across all transformer layers to prevent overfitting. For non-transformer baselines, such as CNNs or RNNs, we use GloVe embeddings pretrained on Wikipedia 2014 and Gigaword 5, with an embedding dimension of 300. For hierarchical classification, we adopt both flat and hierarchical loss functions. In particular, we employ a hierarchical cross-entropy loss that considers the depth-wise error at each level of the taxonomy. In multi-label settings, binary cross-entropy is used with sigmoid activation on the output layer. We also explore contrastive learning for datasets with domain information, where contrastive pairs are generated using domain similarity. In those cases, we jointly optimize a classification loss and a contrastive InfoNCE loss with a coefficient of 0.2. For SciHTC, we implement label smoothing with $\epsilon = 0.1$ to address label sparsity in deep hierarchies. Evaluation is conducted using multiple metrics. For single-label tasks, we report accuracy, macro-F1, and weighted-F1. For multi-label tasks, we evaluate using micro-F1, macro-F1, precision@k, and nDCG@k, with $k = 3$ and $k = 5$. In hierarchical scenarios, we compute hierarchical precision, recall, and F1 using the method introduced in prior work. To ensure fair comparisons, all results are averaged over three independent runs with different random seeds (42, 7, 2023), and we report both the mean and standard deviation. Hyperparameters are selected via grid search on the validation set. Key parameters include learning rate $\{1e^{-5}, 2e^{-5}, 5e^{-5}\}$, batch size $\{16, 32\}$, and dropout $\{0.1, 0.3\}$. We implement all models with mixed-precision training using NVIDIA's Apex to reduce memory usage and accelerate convergence. For CreditRisk, due to class imbalance, we apply focal loss and also experiment with SMOTE for oversampling the minority class. All code is implemented in Python 3.10 with PyTorch 2.1, Transformers 4.36, and scikit-learn 1.3. In addition to standard NLP benchmarks such as THUCNews and SciHTC—which serve

to assess the model's generalization ability across text classification tasks—we further evaluate our approach on two domain-specific datasets that are directly aligned with credit scoring: the German Credit dataset and the Give Me Some Credit dataset from Kaggle. These datasets provide structured tabular data representative of real-world financial credit scenarios. German Credit includes 1000 instances labeled as “good” or “bad” credit risks, with categorical and numerical features. The Give Me Some Credit dataset contains over 150,000 entries including delinquency and credit usage metrics, offering a large-scale benchmark to test model scalability and robustness in practical risk prediction tasks. All datasets are preprocessed via normalization and, where appropriate, categorical encoding. To maintain consistency across evaluations, we apply the same data partitioning strategy (80/10/10 for train/validation/test) and use identical hyperparameter tuning grids. Model performance is measured in terms of Accuracy, Precision, F1 Score, and AUC.

C. COMPARISON WITH SOTA METHODS

In this section, we present a comparison of our proposed approach with several state-of-the-art (SOTA) methods on four benchmark datasets: THUCNews, SciHTC, UTCD, and CreditRisk. We assess the models using various performance indicators, such as accuracy, precision, F1 score, and AUC. The outcomes in Table 1 and Table 2 reveal that our method consistently outperforms existing SOTA models across all datasets. On the THUCNews and SciHTC datasets, our method delivers superior results across all metrics when compared to previous models, including TextCNN [39], FastText [40], BiLSTM [41], Transformer [42], BERT [43], and RoBERTa [44]. On the THUCNews dataset, our approach achieves an accuracy of 94.86 ± 0.02 , significantly surpassing the next best model, RoBERTa, which reaches 92.37 ± 0.01 . Likewise, on SciHTC, our method outshines RoBERTa by over 2 percentage points in both accuracy (90.74 ± 0.02 vs 88.14 ± 0.01) and F1 score (88.93 ± 0.02 vs 86.80 ± 0.02). These results clearly demonstrate the efficacy of our approach in addressing large-scale, multi-class, and hierarchical text classification challenges.

On the UTCD and CreditRisk datasets, our method also exhibits superior performance. On the UTCD dataset, our model reaches an accuracy of 94.16 ± 0.02 , surpassing RoBERTa's performance of 92.03 ± 0.01 . In terms of AUC, we achieve a score of 95.48 ± 0.02 , outperforming all competitors. For the CreditRisk dataset, our model achieves an accuracy of 85.14 ± 0.02 , higher than RoBERTa's 83.49 ± 0.03 . Our model significantly improves the precision and F1 scores, showcasing its ability to handle imbalanced datasets efficiently. The findings suggest that the strong performance of our model is attributed to its enhanced ability to capture contextual information and generalize across multiple domains. Our application of advanced attention mechanisms, contrastive learning, and hierarchical loss functions helps in better capturing long-range dependencies and multi-level label structures, making our model highly

TABLE 1. Performance benchmarking of our approach against leading techniques on THUCNews and SciHTC datasets.

Model	THUCNews Dataset				SciHTC Dataset			
	Accuracy	Precision	F1 Score	AUC	Accuracy	Precision	F1 Score	AUC
TextCNN [39]	89.32±0.03	87.45±0.02	86.90±0.02	90.12±0.02	84.23±0.02	82.10±0.02	82.76±0.01	86.45±0.03
FastText [40]	88.75±0.02	85.91±0.02	85.34±0.03	89.64±0.03	83.59±0.03	80.83±0.02	81.40±0.02	85.32±0.02
BiLSTM [41]	90.18±0.01	88.20±0.03	87.56±0.02	91.07±0.03	85.71±0.02	84.02±0.01	83.91±0.02	87.89±0.02
Transformer [42]	91.03±0.03	88.97±0.02	89.25±0.02	92.33±0.02	86.94±0.01	85.76±0.02	85.91±0.02	88.64±0.01
BERT [43]	92.12±0.02	90.18±0.03	89.74±0.02	93.21±0.02	87.83±0.02	86.52±0.02	86.63±0.01	89.73±0.03
RoBERTa [44]	92.37±0.01	90.43±0.02	90.12±0.03	93.67±0.02	88.14±0.01	87.01±0.02	86.80±0.02	90.01±0.02
Ours	94.86±0.02	92.55±0.02	92.30±0.02	95.42±0.03	90.74±0.02	89.12±0.01	88.93±0.02	92.10±0.02

TABLE 2. Performance benchmarking of our approach against leading techniques on UTCD and CreditRisk datasets.

Model	UTCD Dataset				CreditRisk Dataset			
	Accuracy	Precision	F1 Score	AUC	Accuracy	Precision	F1 Score	AUC
TextCNN [39]	87.29±0.03	84.12±0.02	85.43±0.03	88.77±0.02	77.22±0.03	74.10±0.02	75.05±0.01	81.34±0.02
FastText [40]	86.54±0.01	83.97±0.03	84.21±0.02	88.43±0.03	78.35±0.02	75.93±0.02	74.87±0.03	81.12±0.03
BiLSTM [41]	89.01±0.02	86.34±0.01	87.72±0.02	91.22±0.02	80.47±0.03	77.54±0.02	78.43±0.02	83.98±0.01
Transformer [42]	90.29±0.02	88.21±0.01	89.11±0.03	92.11±0.02	81.56±0.02	78.33±0.02	79.72±0.01	84.45±0.03
BERT [43]	91.76±0.03	89.82±0.02	90.09±0.01	93.56±0.01	82.23±0.02	79.44±0.01	80.01±0.02	85.17±0.02
RoBERTa [44]	92.03±0.01	90.65±0.02	91.33±0.01	94.32±0.02	83.49±0.03	80.27±0.02	80.85±0.03	86.14±0.01
Ours	94.16±0.02	92.04±0.03	91.85±0.02	95.48±0.02	85.14±0.02	82.88±0.01	83.56±0.02	88.52±0.03

effective for both large-scale and hierarchical classification tasks.

Table 3 presents the performance of our proposed model against a suite of strong baselines on the German Credit and Give Me Some Credit datasets. We observe that our model consistently outperforms traditional classifiers such as Logistic Regression and Random Forests, as well as advanced boosting-based models like XGBoost and LightGBM. Specifically, our method achieves an AUC of 85.30% on the German Credit dataset and 90.56% on the Give Me Some Credit dataset, which are both significant improvements over the best-performing baseline (LightGBM). In addition, our model exhibits higher F1 scores, indicating its effectiveness in dealing with class imbalance, which is critical in real-world credit risk assessment. These results validate the domain applicability of our approach. While general NLP datasets demonstrated the model's ability to extract semantic features effectively, the credit scoring datasets further confirm its adaptability to structured financial data, ensuring relevance to practical deployment scenarios.

To enhance transparency and reproducibility, we provide an ablation analysis in Table 4 to quantify the impact of key modeling components. Beginning with a baseline model, we incrementally add structured tuning (grid search, Bayesian optimization), data balancing (SMOTE), and a fairness-aware loss function. Results show progressive gains

in performance and a clear reduction in the Equal Opportunity Difference (EOD), reflecting improved predictive accuracy and fairness. The final configuration achieves the best trade-off between effectiveness and equity.

D. ABLATION STUDY

To further evaluate the effectiveness of each key component in our proposed framework, we conduct a series of ablation studies on all four datasets: THUCNews, SciHTC, UTCD, and CreditRisk. From the THUCNews and SciHTC results in Table 5 and Table 6, we observe a consistent drop in performance when any of the components are removed. For instance, removing the Feature Transformation (Ours w/o Feature Transformation) leads to a drop in F1 score from 92.30±0.02 to 90.92±0.01 on THUCNews and from 88.93±0.02 to 86.04±0.02 on SciHTC. Similarly, excluding the Deep Learning Architecture results in an even greater decrease, particularly on SciHTC, where the hierarchical label structure plays a crucial role. The performance degradation validates the necessity of considering hierarchical dependencies in label space. Moreover, the Regularization and Sampling also proves essential, as its removal leads to a notable reduction in AUC and F1 scores across both datasets, indicating its contribution to domain-aware semantic separation and representation learning.

TABLE 3. Performance benchmarking on credit scoring related datasets.

Model	German Credit Dataset				Give Me Some Credit Dataset			
	Accuracy	Precision	F1 Score	AUC	Accuracy	Precision	F1 Score	AUC
Logistic Regression	76.45±0.03	74.20±0.02	75.03±0.02	78.41±0.03	81.32±0.02	80.11±0.02	80.50±0.02	83.25±0.02
Random Forest	78.21±0.02	76.67±0.03	77.10±0.02	80.93±0.02	83.14±0.02	82.32±0.02	82.44±0.02	85.60±0.02
XGBoost	79.33±0.02	77.44±0.02	78.20±0.02	82.15±0.02	85.76±0.02	84.11±0.01	84.54±0.02	87.90±0.02
LightGBM	80.01±0.01	78.32±0.02	78.94±0.02	83.07±0.01	86.33±0.01	85.02±0.02	85.45±0.02	88.24±0.01
Ours	82.12±0.02	80.41±0.01	81.26±0.02	85.30±0.02	88.74±0.02	87.19±0.01	87.85±0.02	90.56±0.02

TABLE 4. Ablation study on hyperparameter tuning, SMOTE Sampling, and Fairness regularization.

Configuration	Accuracy	F1 Score	AUC	EOD ↓	Notes
Baseline (no tuning, no SMOTE, no fairness)	85.14±0.03	83.07±0.02	87.56±0.02	0.124	Default training
Grid Search tuning only	87.92±0.02	85.81±0.02	89.34±0.02	0.097	Static search over learning rate, dropout
Bayesian tuning only	88.45±0.01	86.52±0.01	90.02±0.01	0.082	Gaussian process guided tuning
+ SMOTE sampling	89.76±0.02	88.14±0.02	91.85±0.01	0.068	Oversampled minority class
+ Fairness-aware regularization	90.74±0.01	88.93±0.01	92.10±0.01	0.036	Adds fairness loss penalty

TABLE 5. Performance benchmarking of our approach against leading techniques on THUCNews and SciHTC datasets.

Model	THUCNews Dataset				SciHTC Dataset			
	Accuracy	Precision	F1 Score	AUC	Accuracy	Precision	F1 Score	AUC
w/o Feature Transformation	92.17±0.02	89.01±0.03	89.43±0.02	92.65±0.01	87.01±0.02	83.94±0.02	84.24±0.03	88.02±0.03
w/o Deep Learning Architecture	93.07±0.01	90.02±0.03	90.32±0.02	93.50±0.02	88.34±0.01	84.67±0.02	85.41±0.01	89.23±0.02
w/o Regularization and Sampling	93.80±0.03	91.12±0.01	90.92±0.01	94.15±0.02	89.51±0.02	85.23±0.01	86.04±0.02	90.14±0.02
Ours	94.86±0.02	92.55±0.02	92.30±0.02	95.42±0.03	90.74±0.02	89.12±0.01	88.93±0.02	92.10±0.02

TABLE 6. Performance benchmarking of our approach against leading techniques on UTCD and CreditRisk datasets.

Model	UTCD Dataset				CreditRisk Dataset			
	Accuracy	Precision	F1 Score	AUC	Accuracy	Precision	F1 Score	AUC
w/o Feature Transformation	90.08±0.01	88.21±0.03	88.56±0.02	91.62±0.01	79.85±0.02	76.29±0.01	77.10±0.02	83.21±0.01
w/o Deep Learning Architecture	90.96±0.02	88.54±0.01	89.04±0.03	92.11±0.02	80.93±0.02	77.52±0.01	78.09±0.03	84.06±0.02
w/o Regularization and Sampling	91.24±0.01	89.03±0.02	89.24±0.02	92.66±0.02	81.63±0.02	78.72±0.03	79.35±0.01	84.79±0.01
Ours	94.16±0.02	92.04±0.03	91.85±0.02	95.48±0.02	85.14±0.02	82.88±0.01	83.56±0.02	88.52±0.03

A similar trend is observed on the UTCD and CreditRisk datasets. The full model achieves the highest F1 score and AUC across both datasets. Notably, on the CreditRisk dataset, the F1 score drops from 83.56 ± 0.02 to 79.35 ± 0.01 when the Feature Transformation module is removed, underscoring its importance in dealing with imbalanced, feature-sparse data. The removal of the Deep Learning Architecture also leads to significant performance degradation on UTCD, which emphasizes its hierarchical taxonomy structure. The Regularization and Sampling module, while slightly less critical than the other two, still provides consistent performance gains,

particularly improving the AUC metric, which is vital in applications such as risk modeling. The ablation results conclusively demonstrate that each of the three core modules—Feature Transformation, Deep Learning Architecture, and Regularization and Sampling—plays a distinct and complementary role in enhancing model performance. The Feature Transformation module facilitates better domain separation and class discrimination, the Deep Learning Architecture strengthens the model's understanding of label relationships, and the Regularization and Sampling mechanism improves semantic alignment and representation precision. Together,

they contribute to a substantial performance boost across diverse datasets and classification settings.

E. FAIRNESS CASE STUDY: DEMOGRAPHIC SUBGROUP EVALUATION

To concretely assess the fairness of our proposed model, we conducted a case study based on a subset of the Give Me Some Credit dataset with demographic annotations (e.g., age and marital status groups). Specifically, we partitioned the test data into two representative subgroups: individuals below and above the age of 40. We then evaluated model performance separately on each group using group-wise F1-score and Equal Opportunity Difference (EOD), a standard fairness metric measuring disparities in true positive rates between groups. Results indicate that our model achieves comparable performance across both subgroups, with F1-scores of 0.882 (under 40) and 0.874 (over 40), and an EOD of 0.036—suggesting minimal fairness gaps in positive class recognition. To mitigate any residual imbalance, we incorporated an additional fairness-aware regularization term in the loss function that penalizes inter-group performance variance during training. This case study demonstrates our framework's ability to maintain equitable predictive accuracy across demographic segments, thereby reducing the risk of unintended bias propagation in real-world applications.

V. CONCLUSION AND FUTURE WORK

In this study, we aimed to enhance scoring prediction by addressing the limitations of traditional statistical models, which often fall short in capturing non-linear and complex relationships in financial data. To tackle this, we proposed a novel deep learning framework that effectively integrates structured financial information with unstructured behavioral data. The model architecture features a hybrid design combining Long Short-Term Memory (LSTM) networks—well-suited for processing temporal sequences—with fully connected layers to model high-level feature interactions. Our pipeline involves an advanced feature transformation process, aggregating statistical, temporal, and financial metrics to enrich the data representation. To ensure robust generalization, we applied adaptive regularization techniques, adjusting the regularization strength dynamically. We also addressed the prevalent issue of class imbalance through dynamic sampling and weighting. Experimental results on benchmark datasets validated the superiority of our approach in terms of both predictive accuracy and interpretability, highlighting its applicability to real-world financial risk assessment.

Despite these promising results, there are still limitations to be addressed. The model's reliance on high-quality and behavioral data could restrict its scalability across institutions with varying data availability. Future work could explore transfer learning or domain adaptation techniques to generalize the model more effectively across heterogeneous data sources. Second, although our architecture improves interpretability relative to black-box models, the explainability of deep learning predictions remains limited compared

to traditional credit scoring models. Further research could integrate attention mechanisms or post-hoc interpretability tools to enhance transparency. These directions aim to broaden the practical adoption of deep learning in financial decision-making systems.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

Conceptual design, Xi Shi; methodology development, Xi Shi; software implementation, Xi Shi; validation procedures, Xi Shi; formal data analysis, Dingfen Tang; investigative work, Dingfen Tang; dataset management, Dingfen Tang; initial draft writing, Dingfen Tang, Xi Shi, Yike Yu; manuscript revision and editing, Yike Yu; data visualization, Yike Yu; project supervision, Yike Yu; funding support, Yike Yu. All contributors have reviewed and approved the final version of the manuscript.

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